Muskan Ejaz

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SUMMARY

Computer Science student at National University of Sciences and Technology with hands-on experience in AI, full-stack web development, and data structures & algorithms. Completed five academic projects, including dysarthria detection (87% accuracy), a real-time water monitoring web app (50+ users), a graph-based social network backend, a hybrid encryption system, and a skill-bartering platform. Skilled in Python, C++, JavaScript, SQL, and Azure. Seeking a software engineering internship for 2026.

TECHNICAL SKILLS

- **Programming Languages:** C++, Python, Java, JavaScript, HTML/CSS, SQL, PL/SQL.
- **Core Concepts:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Systems, Artificial Intelligence, Computer Networks, Information Security, Digital Logic Design, Relational Algebra.
- **Frameworks/Libraries:** NumPy, Pandas, Scikit-learn, Librosa, PyTorch.
- **Web Development:** REST APIs, Node.js, Express.js, React.js, MongoDB, PostgreSQL, API Integration, Frontend Development, Backend Development.
- **Development & Cloud Tools:** Visual Studio Code, Git, GitHub, Azure Identity & Access Management SC-300 Labs, Version Control, Debugging, Team Collaboration.

PROJECTS

Campus Opportunity Aggregator

- Developed Oracle PL/SQL database layer including stored procedures, triggers, and views for personalized recommendations and real-time analytics tracking.
- Built Admin Dashboard using React.js and Node.js/Express REST APIs to manage student opportunities with live statistics, expiring-soon alerts, and interest-based recommendations.

Detecting Dysarthria

- Built an AI classification model using Python, Librosa, and Scikit-learn on 150+ voice samples from UASpeech dataset to identify dysarthric speech.
- Achieved 87% accuracy: reduced feature extraction time by 30%.

Social Graph Explorer

- Implemented BFS, DFS, and Dijkstra's algorithm in C++ to manage 10,000+ user nodes in a social media backend simulation.
- Optimized friend recommendation latency by 40% using adjacency lists, processing 500+ friend requests per second.
- Designed to demonstrate graph algorithms on large-scale social network data.

Quantum Safe Disk Encryption

- Designed a hybrid encryption demo using AES-256 (128-bit key) and Kyber-style KEM for post-quantum key protection.
- Demonstrated secure key recovery mechanism; documented the entire architecture for educational purposes.
- Encrypted and decrypted 1MB+ test files with successful key recovery in under 100ms.

2 Person Video Calling

- Built a real-time two-person video calling app using JavaScript and WebRTC for browser-based communication.
- Implemented peer-to-peer call flow and tested real-time communication between two users.

WORKSHOPS & CERTIFICATIONS

- Agentic AI workshop — Explored AI agent architectures and LLM integration.
- From Code to Cloud — Learned cloud deployment and CI/CD fundamentals.
- SC 300 Labs: Azure Identity and Access Management — Completed hands-on labs for identity and access solutions.
- Micro1 Certificate: Cleared AI interview with outstanding performance in problem-solving and programming fundamentals.

EDUCATION

National University of Science and Technology (NUST) — Islamabad, Punjab Pakistan
Bachelor Of Computer Science — Nov 2024 – Present